

3.24. The absolute value of the velocity and the equation of the streamlines in a velocity field are given by

$$|\vec{V}| = \sqrt{2y^2 + x^2 + 2xy}$$

$$\psi = y^2 + 2xy = \text{constant}$$

Find u and v .

2- Two source are located at $(x,y)=(-1,0)$ and $(1,0)$ and two sinks are located at $(x,y)=(0,0)$ and $(2,0)$, with intensity $\Lambda=\pi\alpha U_\infty$, in the presence of a uniform flow of speed U_∞ . Calculate:

- The stream function and velocity potential.
- The velocity field and stagnation points.
- The stagnation streamlines.
- If you do this question as a take-home, plot the streamlines with MATLAB; otherwise, skip part (d).

